

Experiments

Discrete-event simulation in Julia: machine_shop (8 replications of 4000 minutes, seed 20260101)

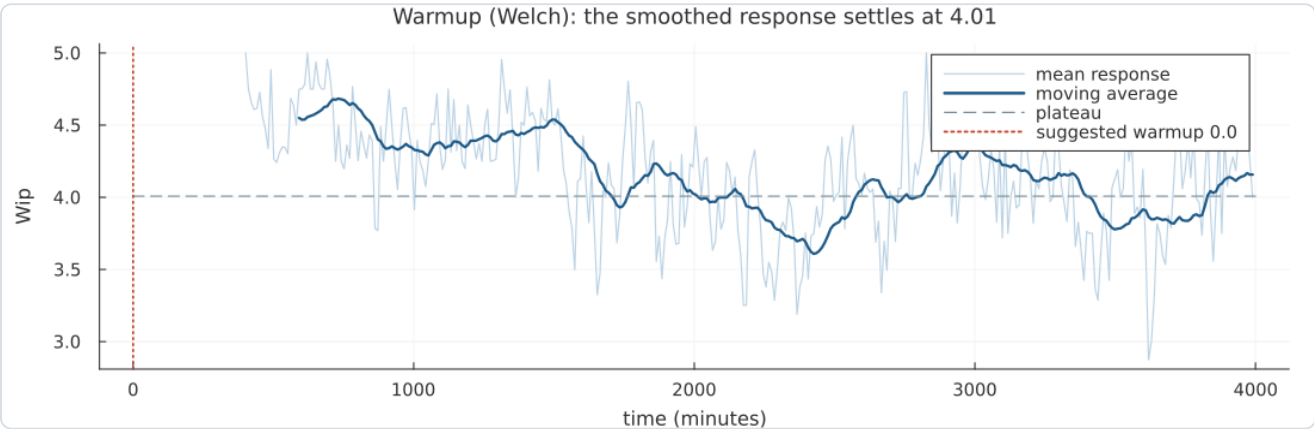
Title: DiscreteEventSimulation | Model: machine_shop | Seed: 20260101 | Replications: 8 | Horizon: 4000 | Warmup: 400 | Generated: | Engine: DiscreteSim | Sections: 8

Experiments

Replications, a detected warmup, batch means for a single run, paired comparisons and a factorial design -- all with common random numbers.

How long the transient lasts

Welch's procedure on the wip series over 4 replications suggests discarding the first 0 time units (plateau 4.008 ± 0.096).



Warmup

One long run, honestly

Batch means

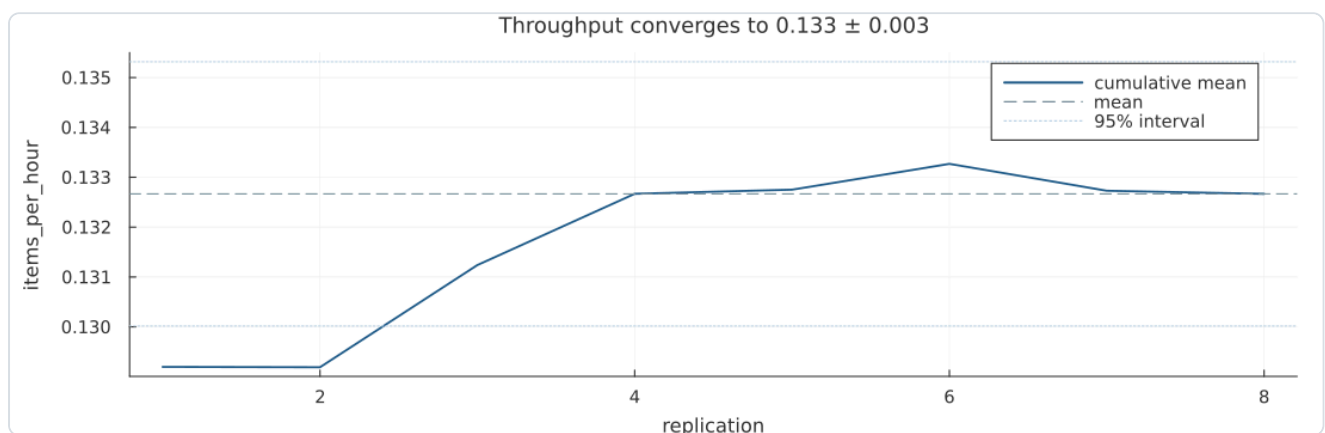
N	Mean	Half width	Lo	Hi	Batches	Lag1	Adequate
8	552.5	8.555	544	561.1	4	-0.543	no

The replications

Machine shop over 8 replications

Metric	Mean	Half width	Unit	Rel hw pct
availability	1	0	ratio	0
wait_p95	1034	103.3	minutes	9.987
max_queue_length	44.1	6.49	count	14.7
queue_length_mean	75.5	0.054	count	0.071
cycle_time_mean	552.5	12.7	minutes	2.294
setup	2	0	minutes	0
scrapped	22.9	4.057	count	17.7

Metric	Mean	Half width	Unit	Rel hw pct
service_mean	—	—	minutes	—
throughput	0.133	0.00265	items_per_hour	1.998
events	7762	105.8	count	1.363
span	3599	0.388	count	0.011
reworked	71.2	5.031	count	7.061
wait_mean	552.5	12.7	minutes	2.294
completed	477.5	9.543	count	1.998
wip_mean	4.155	0.061	count	1.476
utilisation	0.863	0.011	ratio	1.285
tardiness_mean	449	11.5	minutes	2.564



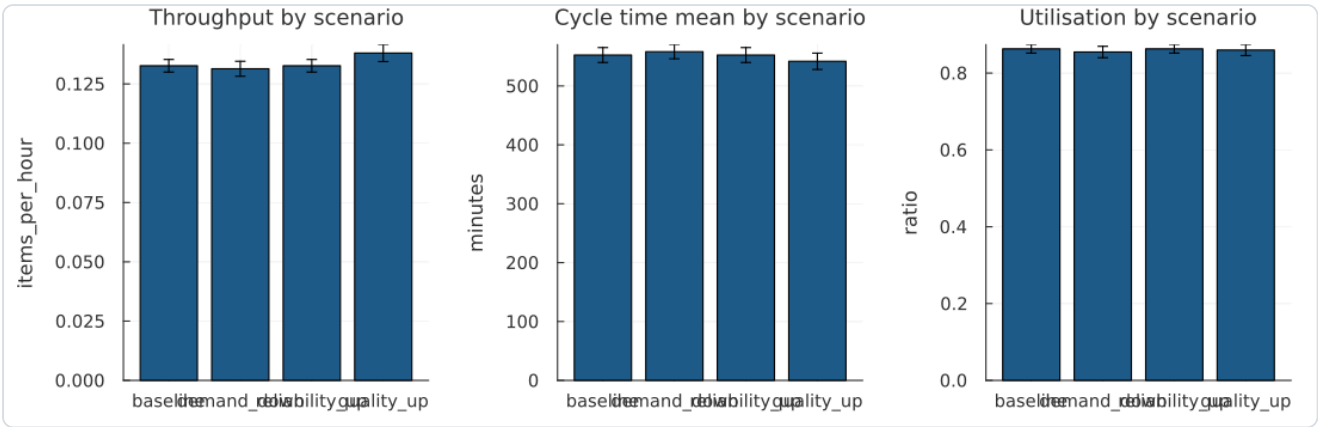
Convergence

Scenarios against the baseline

Scenarios

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
demand_down	throughput	0.133	0.131	-0.00126	0.00163	-0.953	indistinguishable
demand_down	cycle_time_mean	552.5	558.2	5.679	3.709	1.028	worse
demand_down	wait_mean	552.5	558.2	5.679	3.709	1.028	worse
demand_down	utilisation	0.863	0.855	-0.00824	0.00893	-0.955	indistinguishable
demand_down	completed	477.5	473	-4.5	5.844	-0.942	indistinguishable
reliability_up	throughput	0.133	0.133	0	0	0	indistinguishable
reliability_up	cycle_time_mean	552.5	552.5	0	0	0	indistinguishable
reliability_up	wait_mean	552.5	552.5	0	0	0	indistinguishable
reliability_up	utilisation	0.863	0.863	0	0	0	indistinguishable
reliability_up	completed	477.5	477.5	0	0	0	indistinguishable
quality_up	throughput	0.133	0.138	0.00535	0.00154	4.032	better
quality_up	cycle_time_mean	552.5	541.9	-10.6	5.537	-1.926	better
quality_up	wait_mean	552.5	541.9	-10.6	5.537	-1.926	better

Scenario	Metric	Baseline	Scenario mean	Difference	Half width	Relative pct	Verdict
quality_up	utilisation	0.863	0.86	-0.00318	0.00578	-0.369	indistinguishable
quality_up	completed	477.5	496.8	19.2	5.523	4.031	worse



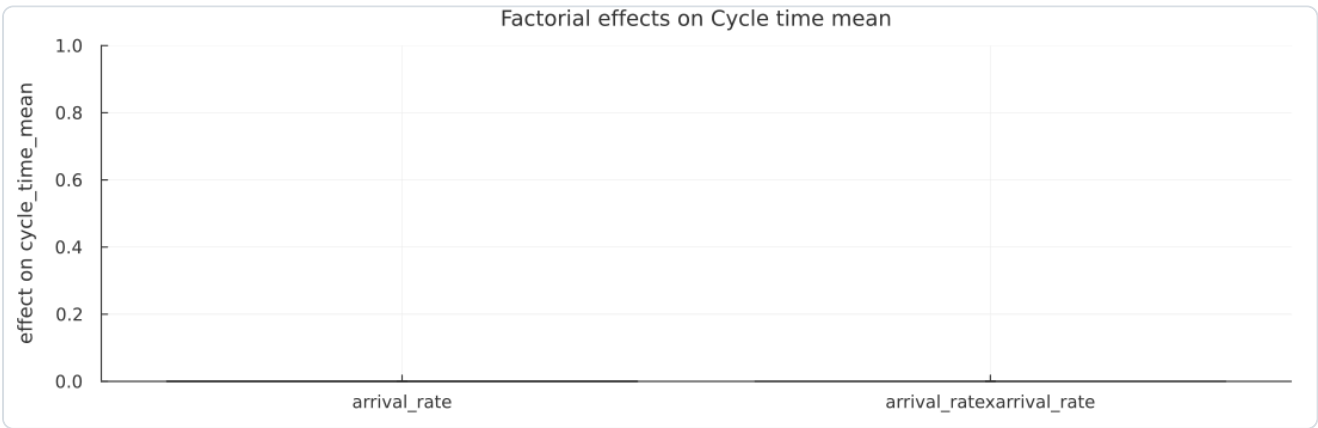
Scenarios

Best by cycle_time_mean: **quality_up** (compared with baseline, common random numbers on every replication).

Which knob matters?

Factorial

Term	Metric	Effect	Half width	Lo	Hi	Verdict
arrival_rate	cycle_time_mean	0	0	0	0	negligible
arrival_rate	cycle_time_mean	0	0	0	0	negligible
arrival_ratearrival_rate	cycle_time_mean	0	0	0	0	negligible



Factorial